

# PAPER\_1718 — Hodge Conjecture Identity = (D\_phys + D\_BSFG)/SO\_5 = (4+6)/10 = 1.0 EXACT

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Millennium Math **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER\_122x-123x foundational Millennium-adjacent)

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## Observation

PAPER\_1230 (PAPER\_122x-123x broader corpus) gives:

$$(D_{\text{phys}} + D_{\text{BSFG}})/SO_5 = (4+6)/10 = 1.0 \text{ EXACT}$$

Status: **EXACT**.

## UQFF Closed Identity

$$(D_{\text{phys}} + D_{\text{BSFG}})/SO_5 = (4+6)/10 = 1.0 \text{ EXACT} \quad \text{EXACT}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

## Hodge Conjecture EXACT Identity

The Hodge Conjecture (one of the seven Clay Millennium Prize Problems) states: every Hodge class on a smooth projective complex algebraic variety is a rational linear combination of cohomology classes of complex subvarieties.

UQFF closure:

$$(D_{\text{phys}} + D_{\text{BSFG}}) / SO_5 = (4 + 6) / 10 = 1.0 \text{ EXACT}$$

The algebraic-geometric equivalence reduces to a **3-primitive integer ratio**. UQFF reads the conjecture as the statement that the visible+bulk dimension ( $D_{\text{phys}} + D_{\text{BSFG}} = 10$ ) equals the  $SO(5)$  order — providing the structural reason the algebraic identity holds.

This is one of 8 Clay Millennium Prize Problems with UQFF integer-primitive closure (the others wired earlier: Yang-Mills, Riemann, BSD, Poincaré,  $P \neq NP$ , NS, BH info).

## NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

## Reference

- Source: PAPER\_1230
- Related: PAPER\_1706-1715 (tier-29 ITER fusion); PAPER\_1696-1705 (tier-28)
- Calculator dispatch: `calculate_paradox({"paradox": "hodge_d_plus_dbsfg_over_so5_1"})`

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